To reduce the redundancy.

⇒ Normalization is the process of organizing data in database. by spiliting large, messy tables into smaller, linked tables to eliminate repeated information and keep data accurate.

The problem: Redundant Data!

Imagine a single table storing customer orders, Every time sarah buys an item, her phone number and address typed out again.

Order ID	customer	Phone	Adress.
101	sarah	555-0192	123 Main St
102	Sarah	555-0192	123 Main st
103	Ali	555-0144	456 Park

→ What is Data Redundancy? and why should we reduce it?

so

→ Repetition of data increases the size of database.

- Other issues like.
  - insertion problems.
  - Deletion problems
  - updation problems.

ex:- St Table.

roll_no	name	branch	hod	office_tel
1	Akon	CSE	Max	552
2	Bkon	CSE	Max	552
3	Ckon	CSE	Max	552
4	Dkon	CSE	Max	552

How Normalization will solve these Problems?
it will break tables into two different tables

Student Table
↓
Student Table + Branch Table

Branch Table

branch	hod	office_tel
cse	Max	552

Students Table

roll no	name	branch
1	Akon	CSE
2	BKon	CSE
3	CKon	CSE

Note:- one can say that ~~Normalization~~ branch name is still Getting repeated in the student table.

So Normalization is not about eleminating the data redundancy. It's about the minimize the data redundancy.

So Now if we want to update a hod and office_tel then it will automatically upaded for all students

→ insertion problem solved.
if we want to insert new student data, we only enter the rollno, the st-name, and just the branch name, because the branch information is stored saperatly. and we dont need to Modify or insert it again.

→ Deletion problem solved.
so when we want to dlete student information. to store the information of the student for the new batch. we just need to deleted all the students record. and we have still Branch Table information is separate table.

→ updation problem solved!
So if we have to update the name of the hod and office-tel no we just need to change in one place

So this was the just Basic concept of Normalization